SCUDEM XI 2026, Problem A: data pack

Pulled on 4 November 2026. All files are CSV.

Files

File	What it is	Feeds
`ghidini_socket_episodes.csv`	273 socket episodes from the Ghidini casefiles. Each row runs from one new prosthesis to its first growth problem (event = 1) or to the next prosthesis or last visit without one (event = 0, censored)	Validation: Kaplan–Meier target for T*
`ghidini_demographics.csv`	62 casefiles: limb-loss level, cause (trauma, congenital, other), age at amputation	Context; no fitting ages
`snyder1977_growth_by_age_sex.csv`	Mean, SD and n by age (2–18) and sex for 7 measurements from 3,900 US children	Growth ODE fits, arm and leg

1. Ghidini casefiles (validation)

Source: Ghidini & Edgar, Data related to 62 Clinical Casefiles of Children with Lower Limb Loss/Absence from Cambodia, Zenodo, doi:10.5281/zenodo.12805856 (open). The raw file is `Casefile Clinical Notes Data.xlsx`: about 820 dated visits with the reason, the clinical evaluation and the intervention.

How the episodes were built

- An episode starts at every visit where a new prosthesis is provided.
- **Event** = the first later visit coded "Growth" with "socket tight" and/or "short".
  - `tight` maps to the model's **tightness criterion**.
  - `short` maps to the **length-discrepancy criterion**.
- **Censored** = the next new prosthesis for a non-growth reason, or the child's last visit.
- Zero-length episodes (two prostheses on the same day) were dropped.

Columns: `level`, `days`, `months`, `event`, `cause` (tight, short, tight+short), `action` (adjustment or new_prosthesis).

Headline numbers (all levels, n = 273 episodes, 155 events)

- Kaplan–Meier median time from a new prosthesis to the first growth problem: **11.8 months**.
- Share still free of growth problems at 12 months: **48%**.
- Event causes: tight 73, short 51, both 31. Of these, 98 were solved with a new prosthesis and 57 with an adjustment.

Limitations

- **No dates of birth or fitting ages.** The demographics give only age at amputation, and only for trauma cases. So T* cannot be matched to a_0 child by child. **Compare the distributions** instead: the model's T* over a plausible range of a_0 against the Kaplan–Meier curve.
- Distal clearance isn't recorded as its own problem. Distal discomfort may be coded as "socket tight" or under "Pain".
- All of this is leg data (transtibial, knee disarticulation, transfemoral). Use it for the leg validation run only.

Cross-check against the paper (PMC12572713): the paper reports adjustments every 8.3 ± 6.6 months and replacements every 16.4 ± 9.9 months. Our numbers use a different definition (time to the first growth problem), so they shouldn't match exactly.

2. Snyder 1977 child anthropometry (growth curves)

Source: Snyder et al. (1977), Anthropometry of Infants, Children, and Youths to Age 18 for Product Safety Design, UMTRI. Individual-level data is hosted by NIST AnthroKids: https://math.nist.gov/~SRessler/anthrokids/orig77/individuals.csv (raw file 1.4 MB, 3,900 children, 123 columns).

Measurements extracted (raw values are in mm and 0.1 kg; converted to cm and kg; sex 1 = male, confirmed from stature at ages 16–18):

`measure`	Snyder variable	Model use
`forearm_length_radiale_stylion`	RADIALE-STYLION LENGTH	Intact forearm ℓ_fa(a); residual length L = ρ·ℓ_fa
`elbow_hand_length`	ELBOW-HAND LENGTH	ℓ_h (elbow to hand) at fitting
`forearm_circumference`	FOREARM CIRCUMFERENCE	C(a), the circumference growth curve
`upper_arm_length_shoulder_elbow`	SHOULDER-ELBOW LENGTH	ℓ_ua(a), shoulder extension
`body_mass`	WEIGHT	M_b(a), strength proxy
`shank_length_tibiale_minus_sphyrion`	TIBIALE HEIGHT − SPHYRION HEIGHT	Leg validation: shank length
`calf_circumference`	CALF CIRCUMFERENCE	Leg validation: circumference

Notes

- Radius length and shank length were measured on a subset (about 1,240 children), so their groups are smaller (n ≈ 7–53). The other measurements have about 3,800 children (n ≈ 30–157 per group).
- Age 18 groups are small; give them less weight in the fits.
- Shank means plateau with small wiggles at ages 15–18. That's sampling noise, and the logistic fit handles it.
- These are intact-limb values from US children in the 1970s. Residual limbs are scaled by ρ, and the Cambodian population differs. State both as assumptions.

Not available

Source	Status	What to do
Ghidini growth cohort, doi:10.5281/zenodo.14187212 (residual and intact limb lengths and girths, 19 children)	**Restricted.** The record is public but its files need approval	Sign in to Zenodo and click "Request access". The paper (doi:10.1097/PXR.0000000000000483) is on Ovid; try your university library for the regression equations and ρ
Otmani et al. 2023 code and data (gitlab.laas.fr/sotmani/InternationalBiomechanics)	Redirects to a sign-in page, although the paper says it is open	Not needed. The paper's finding (doi:10.1080/23335432.2023.2268686) is that adult tables (Dumas & Wojtusch 2018, de Leva 1996), scaled by segment length, best match child data. Apply them to the Snyder lengths
NHANES body measures	Not pulled	No forearm or shank measurements; Snyder covers everything the model needs. Optional cross-check: https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2017/DataFiles/P_BMX.htm